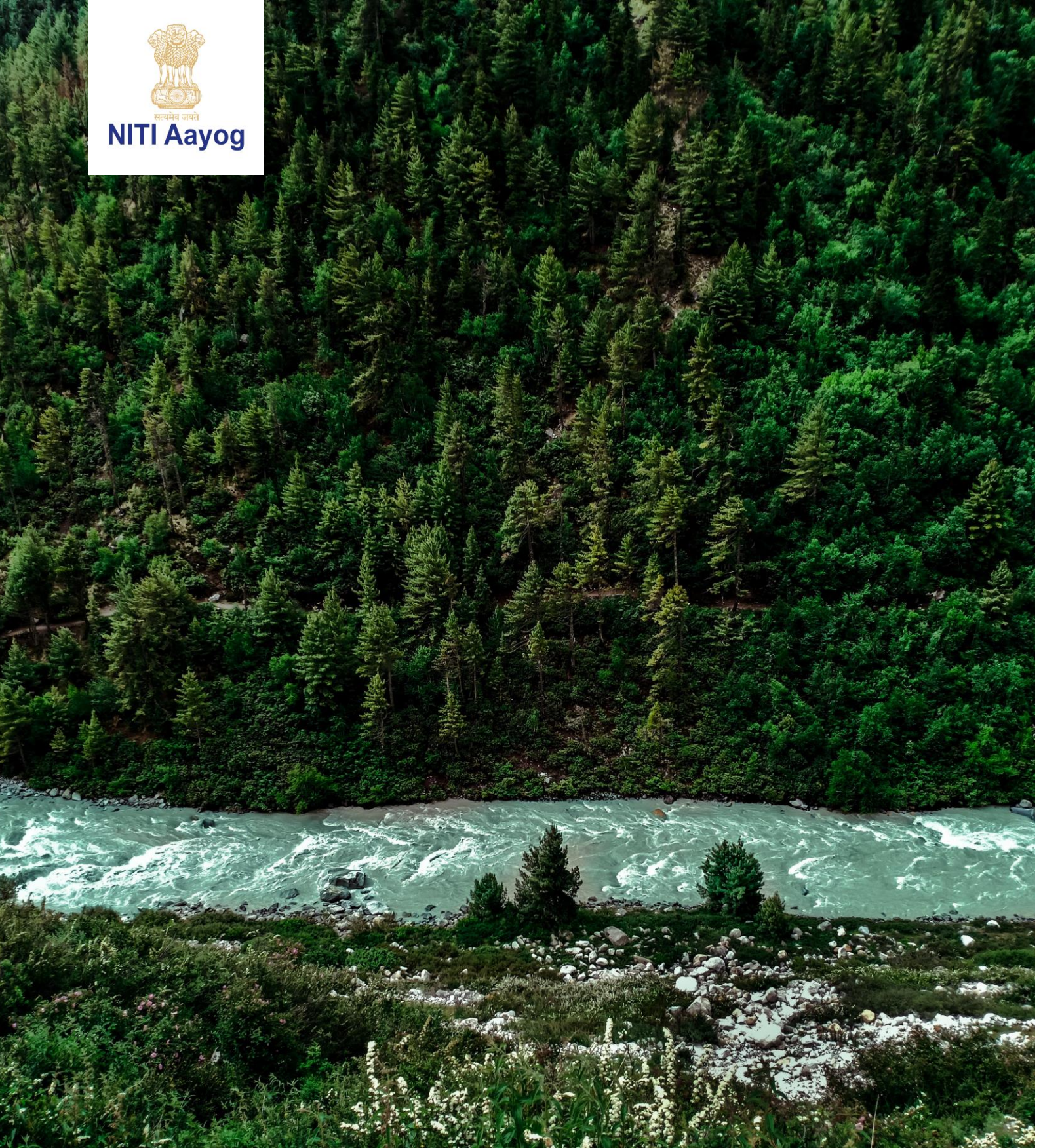




NITI Aayog



SECTOR AT A GLANCE 01

Water Sector in India

An Overview

DRAFT

SECTOR AT A GLANCE 01

WATER SECTOR IN INDIA

AN OVERVIEW

The **'Sector-at-a-Glance'** provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

Suggested Citation:

NITI Aayog (2023). Water Sector in India. NITI Aayog Sector-at-a-Glance Series – Water in India, 2023-Water SAG-01, <link>

Copyright © 2023, NITI Aayog, Government of India

All rights reserved. This draft document and its contents are internal to NITI Aayog. They cannot be reproduced in any manner either in full or parts, thereof. The draft document is subject to further refinements.

All information, ideas, views, opinions, estimates, suggestions, and recommendations (hereinafter 'content') in this draft "portal resource" should neither be understood as professional advice in any manner nor interpreted as policies, objectives, opinions, or suggestions of NITI Aayog. Readers are advised to use their discretion and seek professional advice before taking any action or decision based on these contents. The contents in this portal have been obtained or derived from reliable sources as far as possible. But, NITI Aayog neither represents this information to be accurate nor assumes any liability for any loss, or damages, caused due to any reason whatsoever, towards its usage by a third party. Readers of this document are made aware that the document may be subject to revisions. Any suggestions/input may please be sent to nitiforstates@gov.in.

www.niti.gov.in

Overview

Water in India is hugely important and cuts across agriculture, health, industry, inland transportation, coastal infrastructure, and ecological uses. India has significant water resources - mighty perennial rivers, vast groundwater reserves, large reservoirs, and a long coastline. However, with just 4% of the world's freshwater resources and 18% of the global population, India ranks amongst the most water-stressed nations on a per capita basis. Furthermore, parts of India face a severe and frequent cycle of droughts and floods - thereby affecting agriculture, ecology, industry, habitats, and livelihoods.

The water sector ecosystem is multi-dimensional - comprising policies, programs, infrastructure assets, data monitoring systems, institutions, and community interventions - cutting across Central and State government ministries and departments within water, agriculture, rural development, environment, and several more. National flagship programs like the Atal Bhujal Yojana and Namami Gange are prominent and focus on critical aspects of groundwater and river restoration respectively. A large percentage of States have enacted or are planning prominent programs focused on water such as the AIRBMP in Assam which focuses on managing floods and irrigation or the Jalyukta Shivar Abhiyan 2.0 focused on social and water conservation that was announced earlier in 2023 by the Government of Maharashtra.

Some thrust areas that require policy attention and interventions include efficient water use for agriculture, compliance with water treatment norms in industry, integrated watershed management programs, resilience against flooding and harnessing the blue economy.

For instance, promoting water use efficiency in agriculture through micro-irrigation is critical given farming takes up nearly 90% of freshwater resources in many parts of India. Similarly, ensuring compliance with industrial water use and mandatory wastewater treatment through monitoring systems and incentives or penalties is vital. Scaling decentralized community-driven programs focused on watershed management, groundwater recharge, and catchment treatment under integrated approaches like Jal Shakti Abhiyan will be critical. Expanding the coverage of hyperlocal warning and prediction systems for climate-induced flood disasters and preparing community-level adaptation plans is essential to build resilience in the face of accelerating climate change. Finally, targeted policies and planning frameworks can help unlock the vast blue economy potential across marine fisheries, coastal shipping, tourism and related industries.

Several water issues do not get sufficiently addressed because stakeholders and individuals fail to carry out desired behaviours despite being aware of them. Investments in data and analytics on the one hand and programs that are designed through a scientific behaviour science lens will be critical in shaping progress.

This note covers some high-level trends in various water sub-sectors in India and lists some ideas for the way forward.

Figure 1: Key water statistics

Water resources

690 bcm

Total annual utilizable surface water in India

407 bcm

Total annual extractable groundwater in India

Water use

1486 cubic metres

Average annual per capita water availability for the year 2021
(Per capita water availability of less than 1700 cubic metres is considered water-stressed while that below 1000 is considered water-scarce)

85-90%

of freshwater withdrawals in India goes to the agricultural sector

24.35 lakh hectares

Additional irrigation potential created under the PMKSY between 2016-2022 (against a target of 34.64 lakh hectares by 2026)

Water quality

72%

of urban wastewater was treated as of 2022

54%

of monitored rivers were identified as non-polluted as of 2022

52%

of districts are partly affected by nitrate contamination caused by human activities

2664 million litres/day

of additional sewage treatment plant capacity created through the Namami Gange programme

Climate impacts

75%

Increase in frequency of daily precipitation extremes (rainfall intensities >150 mm per day) between 1950-2015

35%

Increase in instances of extreme and severe floods from 2020 to 2022

Source: PIB, Rising Demand For Water & Efforts For Efficient Management Of Water Resources, 2023; Per Capita Water Availability, 2022; CGWB, Dynamic Ground Water Resources of India, 2023; NITI Aayog, Urban Wastewater Scenario in India, 2022; PIB, Analysis of water quality of rivers by Central Pollution Control Board, 2023; PIB, 24.35 Lakh Additional Irrigation Potential Created Under PMKSY During 2016-2022, 2023; PIB, Impact of Climate Change upon the Indian subcontinent, 2023; PIB, Flash Floods Forecasting, 2023; PIB, Namami Gange Programme, 2023

Sub-sectoral trends

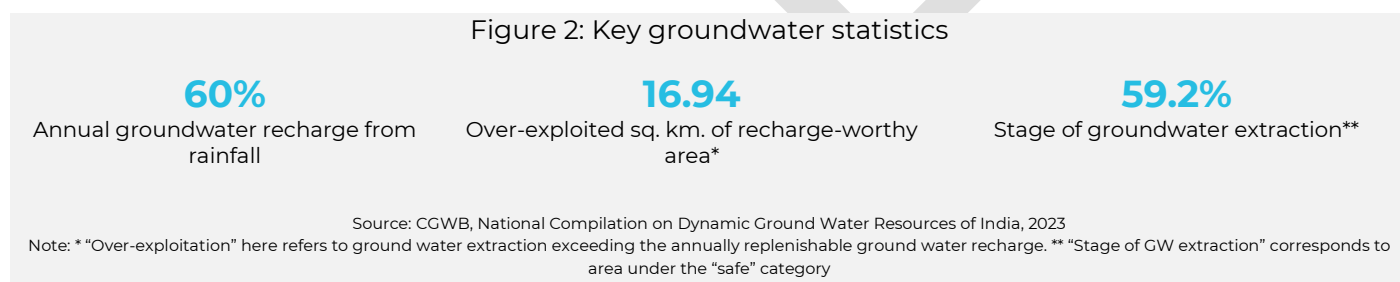
This section presents a summary view of all the relevant statistics and trends related to the following sub-sectors within the water sector:

- Groundwater Management
- Irrigation Water Management
- Industrial Water Management
- Watershed Management
- River Rejuvenation
- Blue Economy
- Water Sharing and Disputes
- Flood Management

Groundwater Management

Groundwater, defined as water stored below ground in aquifers and other sub-surface features, is critical to India. ~40% (447 billion cubic meters or BCM) of India's total utilisable water resource annually (1137 BCM) comes from groundwater.¹

Figure 2: Key groundwater statistics



Groundwater is critical for the well-being of Indians and the sustainability of natural systems:

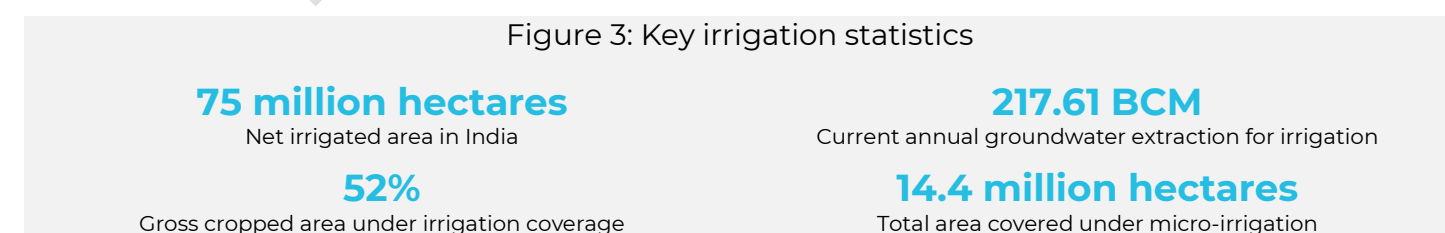
India is heavily reliant on groundwater which contributes 40% of the 1137 BCM of total annual utilizable water resources in India.² It contributes 62% of water for irrigation, 85% to the rural water supply, and 50% to the urban water supply in the country.³

Groundwater occurs in two types of settings- hard-rock aquifers and alluvial aquifers:

Hard-rock aquifers in peninsular India are vulnerable to drying out due to poor permeability. On the other hand, alluvial aquifers in the Indo-Gangetic plains are vulnerable to irreversible overexploitation caused by excessive extraction.⁴ The size of these aquifers is also larger in the Indo-Gangetic plains leading to more pronounced common-pool resource management problems where farmers agree to limit extraction.

Irrigation Water Management

Figure 3: Key irrigation statistics



¹ NITI Aayog, Composite Water Management Index 2.0, 2019

² Ministry of Water Resources, PIB, Government has Taken a Number of Initiatives for River Development and Rejuvenation, 2018

³ Central Ground Water Board (CGWB), Dynamic Ground Water Resources of India, 2022

⁴ PRS, Overview of Groundwater in India, 2016

Irrigation water management, or the management of processes and systems through which water reaches and is applied to farming activities, is a critical subject for India given our focus on agriculture. Effective management of irrigation water directly improves yields, food security, and farmer incomes, thereby playing an integral role in India's economic stability and development.

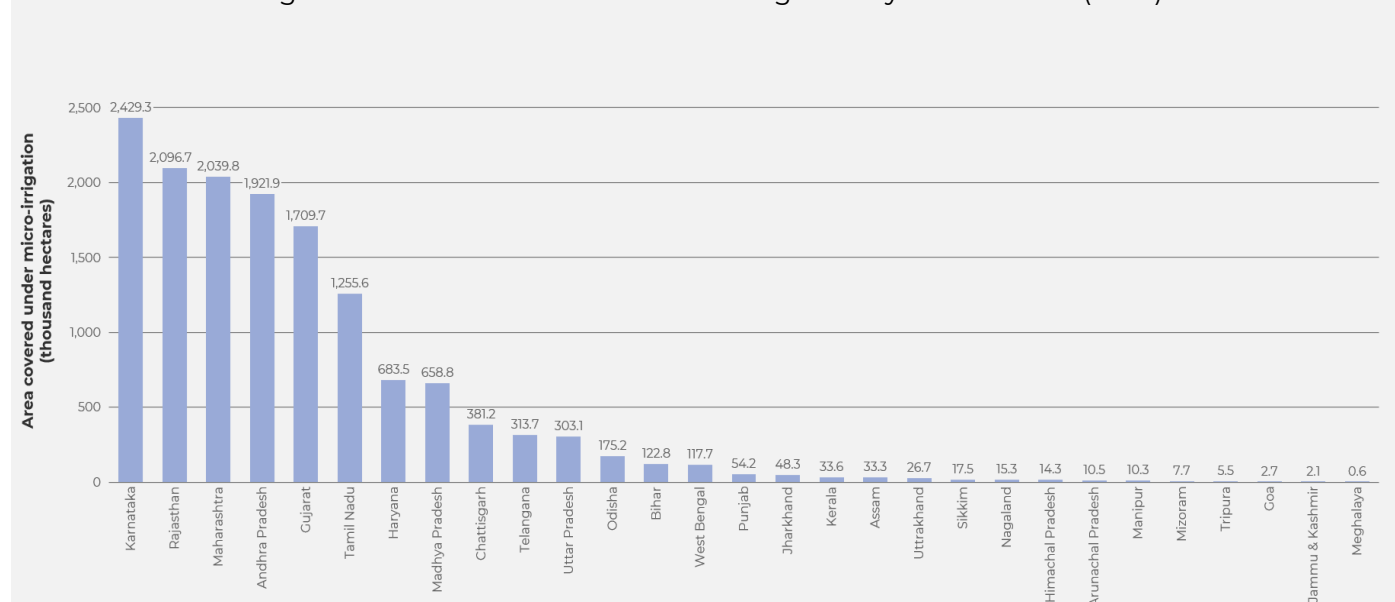
Low water use efficiency (WUE) in irrigation:

The economic WUE in agriculture for India is 0.42 USD/m³ compared to 38.85 USD/m³ and 24.83 USD/m³ for the industrial and services sectors.⁵ Similarly, the economic WUE for agriculture in countries such as Israel (2.12 USD/ m³), China (2.1 USD), and Bangladesh (0.84 USD) is significantly more than in India. Put differently, India utilizes two to three times more water per unit of major food crop production compared to nations like China, Brazil, and the USA.⁶ Some reasons for this are the use of traditional methods of irrigation, especially flood irrigation, poor penetration of micro-irrigation systems, farmer behaviour and mental models, as well as crop selection governed by incentives. Addressing this will be vital going forward.

Absence of irrigation systems results in groundwater extraction:

Due to the partial penetration of formal irrigation systems across the country, more than 50% of Indian agriculture is rainfed and relies on groundwater extraction. This drives severe groundwater depletion in states like Punjab, Haryana, Uttar Pradesh, and Maharashtra.⁷

Figure 4: Area covered under micro-irrigation by state and UT (2022)



Source: Ministry of Agriculture and Farmers Welfare, Agricultural Statistics at a Glance, 2022

Industrial Water Management

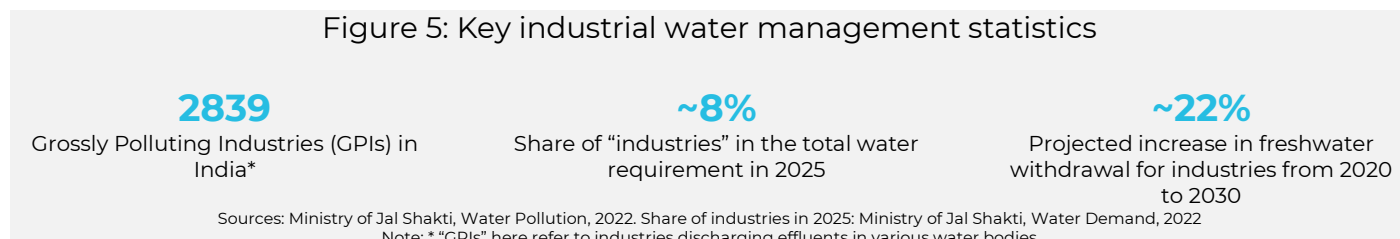
⁵ UN Water, SDG 6 Progress, 2019

⁶ United Nations, The Perfect storm: Pathways to manage India's water sector sustainably, 2018

⁷

Industrial water management involves the efficient and responsible use of water resources through water conservation, reuse, recycling, and wastewater treatment practices. Industrial water management is critical considering the significant industrial usage of water, low water productivity and contribution to water pollution.

Figure 5: Key industrial water management statistics



Excessive withdrawal of freshwater:

Industrial water demand accounts for 7-8% of the total freshwater withdrawal in the country, with low water use efficiency of ~35 USD/ m³, which is less than 65% of countries worldwide.⁸

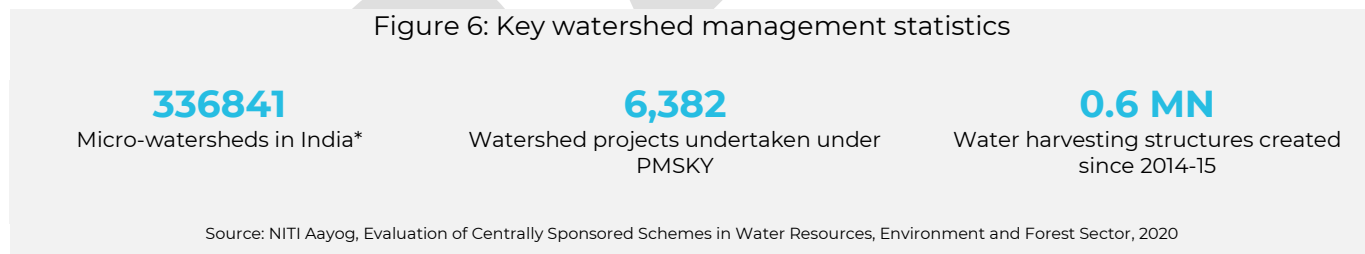
A major contributor to water pollution:

Industrial wastewater and the discharge of untreated sewage are the major causes of water pollution in 323 rivers in India.⁹ A significant fraction of industrial wastewater also contains hazardous chemicals and pollutants that are harmful to human and animal health, and also diminish biodiversity in the region.

Watershed Management

Watershed management refers to the careful management of soil, vegetative cover, and water resources within watershed boundaries to ensure ecological balance, while simultaneously enabling better agriculture and human activity. In India, watershed management has long been recognised as critical and has witnessed progress over the years due to government initiatives such as the Pradhan Mantri Krishi Sinchayee Yojana – Watershed Development (PMKSY-WD) or the erstwhile Integrated Watershed Management Program (IWMP).

Figure 6: Key watershed management statistics



Promotes water security & groundwater recharge:

Watershed management guidelines in India promote the creation of structures such as check dams to arrest water run-offs, and soil erosion and improve water infiltration thereby arresting the fall in groundwater levels, and optimising rainwater harvesting and usage. This is critical for India given groundwater depletion levels.

Drives food security & livelihoods:

Watershed management programs also seek to improve soil health which is critical in India given one-third of the land area in India is affected by different forms of degradation. Effective watershed management is also critical to support rural livelihoods – agriculture, forestry, livestock rearing – something that is vital to more than two-thirds of the population.

Improves climate adaptation & disaster resilience:

⁸ UN Water, Progress on Water-Use Efficiency, 2020 CPCB, Industrial Water Supply; accessed July 2023; the base for calculating the 65% figure countries where change in water use efficiency was tracked in 2020

⁹ CPCB, River stretched for restoration of water quality, 2022

Appropriate watershed management also reduces the impacts of floods through arresting the flow of water and reducing soil run-offs, factors that exacerbate flash flooding. As already discussed, effective watershed management improves soil moisture levels and also groundwater thereby improving drought resilience for rural communities.

River Rejuvenation

River rejuvenation in India refers to the sustainable revitalisation of river ecosystems for economic development, improved livelihoods, and river health through ecological restoration, pollution control, and sustainable water management practices.

Figure 7: Key River Rejuvenation Statistics in India

Estimated benefits from the rejuvenation of 13 major rivers

1900 million m³/yr

Groundwater recharge

50 million tonnes of CO₂e

Additional carbon sequestration in ten years

Source: Indian Council of Forestry Research and Education, Overview of Detailed Project Reports for Rejuvenation of Major Indian Rivers through Forestry Interventions, 2022

Rivers play a critical role in Indian life:

India is home to several key river systems that play a significant role in providing drinking water, irrigation, power generation, transport, fisheries, and tourism. Just the Ganga basin alone is home to 600 million people for instance¹⁰. Ensuring good health and appropriate flows for these rivers is extremely critical for the country.

Almost half the rivers are polluted:

However, the state of Indian rivers needs urgent attention due to challenges in both flow and also water quality. For instance, a 2022 study of 603 rivers by the CPCB revealed 311 polluted stretches in 279 rivers with the highest in the states of Maharashtra, Madhya Pradesh, Karnataka, and Kerala.¹¹

Blue Economy

Blue economy refers to the sustainable and responsible use of ocean and marine resources for economic growth. India's blue economy holds tremendous potential for economic growth and livelihoods given its vast marine territory.

Figure 8: Key blue economy statistics

7,500 km

Coastline length of India

30%

Coastal states in India

4 MN+

Fisherfolk and coastal communities sustained by the coastal economy

95%

India's trade volume transiting by sea

4% of GDP

Estimated size of India's blue economy

Source: Economic Advisory Council to the Prime Minister, India's Blue Economy: Draft Policy Framework, 2020

India's 7500-km-long coastline carries tremendous economic potential:

Indian coastline stretches over 7,500 kilometres across 9 states and 4 UTs encompassing 12 major ports and 200 minor ports. India's blue economy supports 95% of the country's business through transportation and contributes an estimated 4% to its GDP.¹² Fishing, shipping, aquaculture, and port

¹⁰ The World Bank, The National Ganga River Basin Project, 2015

¹¹ Central Pollution Control Board (CPCB), River stretched for restoration of water quality, 2022

¹² TERI, Blue economy: An ocean of livelihood opportunities in India

activities have immense livelihood potential with fishing alone employing 28 million fishers.¹³ These are fast-growing sectors with shipping witnessing a 35% growth in the seafaring workforce in 2017-18.¹⁴

India's fisheries and aquaculture sector is on a high-growth trajectory:

India is currently the 2nd largest fish-producing nation and fish-related activities provide livelihood support to a total 28 million fishers. Out of these, 2.78 million fishers are full-time actively engaged in fisheries and fisheries related activities.¹⁵ This trend is likely to continue as India has been estimated to register a compound annual growth rate of ~25% in fisheries between 2016-30, with aquaculture expected to grow at 44%.¹⁶

Water Sharing and Disputes

Water-sharing efforts involve defining water use rights and maintenance responsibilities, setting up institutions for dispute resolutions, and guidelines on the export and import of water. International and inter-state disputes remain a cause for concern. Ensuring effective water sharing and dispute resolution mechanisms is becoming more important due to depleting water resources.

Figure 9: Key water sharing and distribution statistics



India shares rivers and has signed water-sharing treaties with several other countries:

India shares the water of its major rivers, including the Indus, Ganges, Brahmaputra, Mahakali, and Teesta, with other countries and has enacted treaties with several countries, including Pakistan, Bangladesh, and Nepal, for mutual benefits of water resources.¹⁷

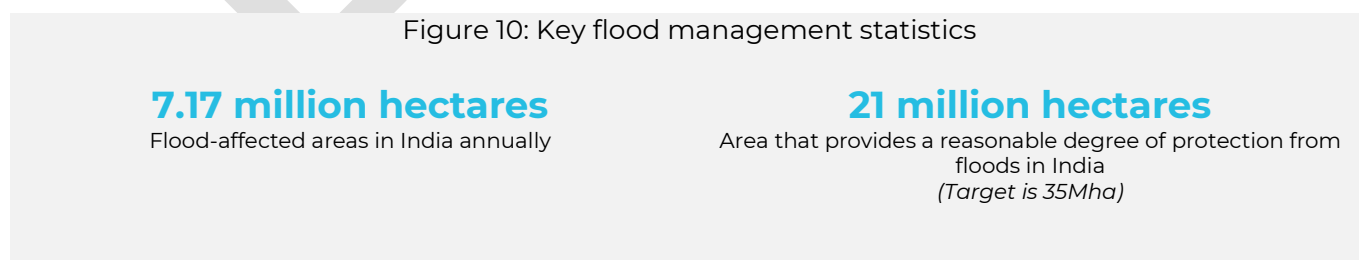
Inter-state tribunals have been developed to solve water-sharing disputes among states:

Many Indian states also share rivers. The Indian government, since 1956, has created inter-state river water tribunals for solving water-sharing disputes for rivers such as Krishna, Cauvery, and Godavari, out of which five were active as of 2018.¹⁸

Flood Management

Flood management refers to the measures and strategies aimed at preventing and mitigating the impact of floods. This includes flood forecasting and warning (FFW) systems, infrastructure development, and disaster response planning. Flood management in India is highly crucial owing to increasing disaster risks and needs a pragmatic and multifaceted approach that balances protection and economic costs.

Figure 10: Key flood management statistics



¹³ Ministry of Fisheries, Fishing communities, 2022

¹⁴ PIB, Employment of Indian seafarers grows by an unprecedented 35 percent this year, 2019

¹⁵ National Fisheries Development Board (NFDB), Fish and Fisheries of India, 2020; Ministry of Fisheries, Fishing communities, 2022

¹⁶ Invest India, India's Blue Economy, 2022

¹⁷ PIB, Water Treaty/Agreement, 2022

¹⁸ PIB, Inter-State River Water Sharing Disputes, 2018

INR 15641 crores

Funds released under various flood management programs

1.7% of GDP

Annual investment needed for adaptation measures,
including for floods

Source: NITI Aayog, Flood Report, 2021; National Disaster Management Authority, National Workshop on Flood Preparedness and Response, 2023; Ministry of Jal Shakti, State-Wise Funds Released under Flood Management Programme during XIth, XIIth Plans & FMABP scheme (2017-22), 2023

Three-fourths of all states and UTs are vulnerable to floods:

75% of Indian states and UTs have been assessed as being vulnerable to floods and other hydrological meteorological hazards and 60% of all Indian districts as extreme flood event hotspots.¹⁹ Though the coverage of forecasting stations in states such as Assam and Bihar is dense, states such as Uttarakhand and Kerala would benefit from increasing the coverage of such stations.²⁰

¹⁹ CEEW, 75% districts and half of India's population vulnerable to extreme climate events, 2020; CEEW, Preparing India for Extreme Climate Events, 2020

²⁰ Standing Committee on Water Resources, Flood Management in the Country including International Water Treaties in the field of Water Resource Management with particular reference to Treaty/Agreement entered into with China, Pakistan and Bhutan, 2021

Opportunities and challenges

Opportunities

 Improvement in agricultural productivity and food security	• Optimal groundwater management for irrigation can lead to an increase in the net benefit of irrigation. • Utilising watershed development programs for developing rainfed areas enhances agriculture productivity. • Growing less water-intensive crops can improve agricultural productivity. • Improved WUE enhances agricultural productivity, thus sustaining food production and ensuring greater food security. • Effective water sharing increases available water for irrigation, doubling agricultural productivity per unit of land.
 Greater economic outcomes	• Utilising watershed programs to bring irrigation facilities and develop water harvesting structures can improve economic outcomes for farmers and reduce their debt burdens. • Greater watershed investments can support poverty alleviation by enhancing employment in rural areas. • Effective flood management can help mitigate huge economic losses caused by floods. • Improvements to water quality, availability, and biodiversity along rivers can boost tourism, health, transportation, food production, and other river-based industries, generating revenue and jobs, especially for rural communities.
 Achievement of water and overall environmental sustainability	• Utilising techniques like micro irrigation which have high water use efficiency enabling conservation of water resources. • Undertaking climate-resilient integrated flood management efforts can help India navigate extreme river floods. • Managing water flows with barrages or dams and limiting extraction during dry periods can raise groundwater levels and water security. • Protecting river habitats can support more diverse and abundant fish and wildlife populations, including endangered species like the Ganges dolphin. • Limiting pollution and encouraging sustainable sourcing and usage of water can support healthier environments, communities, and economies. • Transitioning to alternative sources like treated wastewater or desalination eases the burden on rivers and aquifers.
 Potential for growth in emerging areas	• Investing in innovative solutions for efficiency and monitoring, recycling, and alternative sources can help develop human and technological capabilities for continued progress. • Leveraging the high potential of fisheries and aquaculture which employ over 14 MN people in India can stimulate growth. • Utilising the potential of offshore wind, wave, thermal, solar, and tidal energy sources can boost energy generation. • Focusing on sustainable coastal and marine tourism has high potential in India, especially around islands.

Challenges

 Inadequate waste management efforts	• Discharge of untreated effluents by industries, excessive fertilizer and pesticide usage in agriculture, and improper solid and liquid waste management lead to GW contamination. • Dependence on old technologies which are often outdated and inefficient adversely affects industrial wastewater treatment. • High pollution levels in rivers threaten fish and wildlife in river ecosystems.
 Inefficient water use and related practices and implementation	• Over-reliance on water-intensive crops (like rice and wheat) and inefficient irrigation practices (such as limited use of micro-irrigation systems) lead to excessive GW withdrawals. • Low land sanction rates and incorrect land classification alongside non-adherence to scientific approaches affect the implementation of watershed projects. • Inadequate focus on the maintenance of related structures and defined practices (like regulation of GW extraction) affects the long-term sustainability of natural resources. • Illegal and unscientific sand mining severely threatens river ecology, with subsequent risks of flooding and destruction of natural habitat • Overfishing, pollution, coastal erosion, and habitat destruction threaten fish stocks and the health of coastal and marine ecosystems.
 Limited existence, use, and effectiveness of institutional provisions and efforts	• Lack of coordination and data sharing between concerned bodies, alongside lack of GW legislation and implementation in states affect related measures. • Limited number of states utilising metered charge mechanisms curbs incentives for farmers to reduce GW over-extraction. • Inadequate focus on floodplain restoration in programs and limited state-level floodplain zoning legislations impact river rejuvenation. • Constraint government budget affects development in emerging industries like ocean energy, tourism, mining, ports, and shipping. • Lack of defined time limits for various operations and absence of age-related rules for the appointment of Tribunal members affect the functioning of Tribunals for settling water disputes. • Concerns related to national security, sovereignty, and territoriality bring in complexity to the governance and consequent use of transboundary river water.
 Insufficient capacity building and communication	• Limited farmer consultation as reported by WDC PMKSY's impact evaluation and insufficient training as per a survey on WDC affects the capacity of such stakeholders.²¹ • Lack of awareness of responsibilities and conflict resolution mechanisms amongst WUA members leads to poor operations.

	• Focus of existing forecasting systems on predicting river water levels and display of difficult-to-comprehend, technical information leads to ineffective communication of real flood risk.
 Limited infrastructural facilities	• Limited groundwater recharge infrastructure and reduced potential areas in cities to set up such infrastructure hinder GW recharge. • Ad-hoc methods of setting up flood forecasting stations as per requests from state governments have resulted in unpreparedness during floods. • Slow progress in the rehabilitation of natural drainage systems and scarce pre- and post-monsoon inspection of large dams by states leads to challenges in drainage management.

Institutional framework and policy landscape –

Key ministries and departments

Waterfalls under the State List of the Constitution, give legislative assemblies the power to make laws on the subject within the prescribed guidelines of the centre.

The central government is responsible for the formulation of broad policies and frameworks that guide states in managing their water resources, resolving international and inter-state water conflicts, providing financial assistance to states, and monitoring and regulating water resources in India. At the centre, the Ministry of Jal Shakti is the nodal ministry responsible for the overall planning and policy formulation related to water resources in India. It aims to ensure sustainable and equitable management of water, both in terms of availability and access. It is further bifurcated into two departments with distinct mandates concerning the different uses of and concerns over water resources.

The Department of Water Resources, River Development and Ganga Rejuvenation (DoWR, RD & GR) is the nodal agency for water resource management, responsible for building national planning, perspective, policies and programmes for the development, conservation, and management of water as a national resource. The Department of Drinking Water and Sanitation (DoDWS) provides technical and financial assistance to the States to provide safe and adequate drinking water to rural India with a focus on service delivery.

Central government institutions such as the Central Water Commission and the CPCB are also involved in different aspects of water resource management such as mobilizing and facilitating scientific, and technical inputs and finance across states, regulating water quality, reviewing national policies from time to time, responding to the contemporary challenges to water resources, conducting research and development, and monitoring data for forecasting.

The central government also addresses inter-state water-sharing disputes by establishing tribunals under the ISRWD Act of 1956. Currently, there are five active tribunals. After the resolution of disputes by the tribunal, suitable mechanisms like control authorities, supervisory committees, or boards are established to ensure the implementation of the award. These regulatory organisations such as Godavari River Management Board, and Upper Yamuna River Board function under the DoWR, RD & GR.

At the state level, the respective governments legislate and implement policies based on broad guidelines provided by national frameworks such as the National Water Policy and Model Bills.

Multiple state ministries and departments are involved in water management at the state level as well. Departments of various ministries such as the Department of Water Resources, Department of Agriculture/Horticulture, Department of Rural Development and State Disaster Management Authorities lead efforts on policy and program implementation on groundwater management and irrigation water management. State-level institutions like the State Water Boards, Public Works Department, and the State Pollution Control Board conduct functions such as water quality monitoring, and monitoring of data, preparations and response for hydro-met natural disasters.

Key ministries and departments



Ministry of Jal Shakti

Oversees the management and conservation of water resources in India, ensuring equitable access to safe and clean water while promoting sustainable water practices across various sectors.



Ministry of Agriculture & Farmers' Welfare

Formulates and implements policies and programs to enhance irrigation infrastructure, promote efficient water use in agriculture, develop rainfed areas, and

support farmers in adopting sustainable water management practices for improved agricultural productivity.



Ministry of Rural Development

Formulates and implements policies and programs related to watershed development and undertakes water conservation and storage works under MGNREGS.



Ministry of Environment, Forest & Climate Change

Oversees the conservation and management of natural resources including water.



Central Water Commission

Initiates and coordinates the schemes for water control, conservation, and utilization with state governments.



Central Pollution Control Board

Responsible for prevention, control, and abatement of water pollution in accordance with the Water (Prevention and Control of Pollution) Act, 1974.



Central Ground Water Board

National agency responsible for providing scientific and technical inputs for management, exploration, monitoring, assessment, recharge, and regulation of groundwater resources of the country.



Central Groundwater Authority

Set up in 1997 under the Environment (Protection) Act 1986 for regulation, conservation, protection of groundwater and mass awareness; wide powers including the power to impose a penalty on any person, company or government department.



National Water Resources Council

Lays down the National Water Policy and reviews it periodically, assists in the resolution of inter-state water conflicts.



National Institute of Hydrology

Conducts research and development activities in water resources management; collects and analyses hydrological data, develops forecasting models, and assesses water quality.



National Water Development Agency

Carries out water balance and scientific studies for optimum utilization of water resources of the Peninsular River system.

National Water Informatics Centre



Maintains the central repository of water resources in the country.



National Green Tribunal

Responsible for effective and expeditious disposal of cases relating to environmental protection and conservation of forests and other natural resources



Ganga Flood Control Commission

Responsible for preparation and regular up-gradation of master plans for flood management in the Ganga basin and sub-basins.



Brahmaputra Board

Responsible for planning and integrated implementation of flood control measures in Brahmaputra and Barak basin in northeast India.



Godavari River Management Board

Responsible for equitable sharing of Godavari waters by Andhra Pradesh & Telangana based on relevant awards and resolving disputes.



Upper Yamuna River Board

Responsible for allocation of available flows amongst the beneficiary states and monitoring the return flows; monitoring conserving and upgrading the quality of surface and groundwater and maintaining hydro-meteorological data for the basin.



Krishna River Management Board

Regulates water allocation and distribution among the states in the Krishna River Basin, resolves water disputes, and facilitates sustainable water resource management and planning.

Major policies and programs

This section highlights existing policies related to the water sector and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



National Water Policy, 2012²²

National Water Policy establishes a national framework defining general principles for water governance and devolution of authority.



Atal Bhujal Yojana²³

ABY aims to enhance the sustainable management of groundwater at the community level by:

- Promoting coordination among different Central and State programs
- Bringing community behaviour change through awareness initiatives and skill development, ultimately fostering sustainable groundwater management.

²² Ministry of Jal Shakti, GoI, National Water Policy, 2012

²³ Ministry of Jal Shakti, GoI, Atal Bhujal Yojana, 2020



Namami Gange Programme, 2015²⁴

Namami Gange is an integrated conservation mission for the reduction of pollution, conservation, and rejuvenation of the Ganga River through setting up sewage treatment infrastructure, river-front development, river-surface cleaning, industrial effluent monitoring, afforestation, biodiversity conservation, public awareness, and Ganga gram.



Pradhan Mantri Krishi Sinchayee Yojana, 2015²⁵

PMKSY aims to promote the coordination of investments in irrigation at the individual field level through four sub-components: Accelerated Irrigation Benefit Programme, Har Khet Ko Pani, Per Drop More Crop, Watershed Development Component by:

- Expanding the cultivable land area with reliable access to irrigation
- Improving the efficiency of water usage on farms to minimise wastage and promoting the adoption of precision irrigation and other water-saving technologies
- Enhancing the replenishment of groundwater sources and implementing sustainable water conservation practices



National Water Mission²⁶

NWM aims at the conservation of water, minimisation of wastage, and more equitable distribution of water across and within states through integrated water resources development and management.

²⁴ Ministry of Jal Shakti, Namami Gange, 2015

²⁵ Gol, PMKSY, 2015

²⁶ Ministry of Jal Shakti, Gol, National Water Mission, 2011

Way forward

India's water sector is at a critical inflexion point. Reforms and changes over the last few years have led to a more streamlined governance and institutional structure, the role of data-based decision-making is now increasingly well recognized, and water is now increasingly getting treated as a scarce resource. However, there are significant changes that need to be implemented, urgently, to sustainably manage, perhaps, India's most vital natural resource and one that will determine the fate of hundreds of millions of Indians going forward.

Some areas that require attention and could drive significant value are:

 Enhancing coordination	Greater coordination between activities related to groundwater management, irrigation water management, and watershed development through institutional change management and supporting data will be critical to avoid a siloed approach.
 Enabling better community planning	Helping communities with the adoption of hyperlocal data-based approaches to plan and manage watershed management projects will be essential to realize and sustain gains.
 Scaling innovative irrigation and farming practices	Scaling the use of micro-irrigation programs as well as traditional climate-resilient farming practices through a package approach that combines funding, technical support, and maintenance holds significant potential for increasing water-use efficiency and mitigating climate risks.
 Advancing financial and operational sustainability	Strengthening rules to manage industrial and urban discharge of wastewater into rivers will be essential. Business and financial models should be designed that drive high-performance and self-sustaining operations. Models such as Payment-For-Ecosystem Services (PES) can be looked at in certain locations and certain sectors.
 Leveraging non-traditional capital sources	Tapping into climate finance and other innovative financing mechanisms to build and run water management projects including those linked to blue carbon – mangrove ecosystems, seagrasses, and salt marshes – in coastal areas will jointly advance water conservation and carbon emission goals.



**Accelerating
data-led
decisionmaking**

Elevating the role of data and behavioural science in the management of water and investing in best-in-class climate and water models and decision-making related to these models will be essential for continued high-quality policymaking and implementation.

DRAFT

Recommended readings

NITI Aayog. (2023). *Compendium of Best Practices in Water Management*.

https://mohua.gov.in/upload/uploadfiles/files/Part_A-Chapter-4-Planning-and-Development-of-Water-Sources.pdf

CEEW. (2023). *India's Sustainable Water Management Story*.

<https://www.ceew.in/sites/default/files/managing-indias-water-resources-in-sustainable-equitable-ways.pdf>

NITI Aayog. (2019). *Composite Water Management Index 2.0*.

<https://www.niti.gov.in/sites/default/files/2023-03/Composite%20Water%20Management%20Index%202.0.pdf>

NITI Aayog. (2018). *Strategy for India @75*.

https://www.niti.gov.in/sites/default/files/2019-01/Strategy_for_New_India_2.pdf



www.niti.gov.in